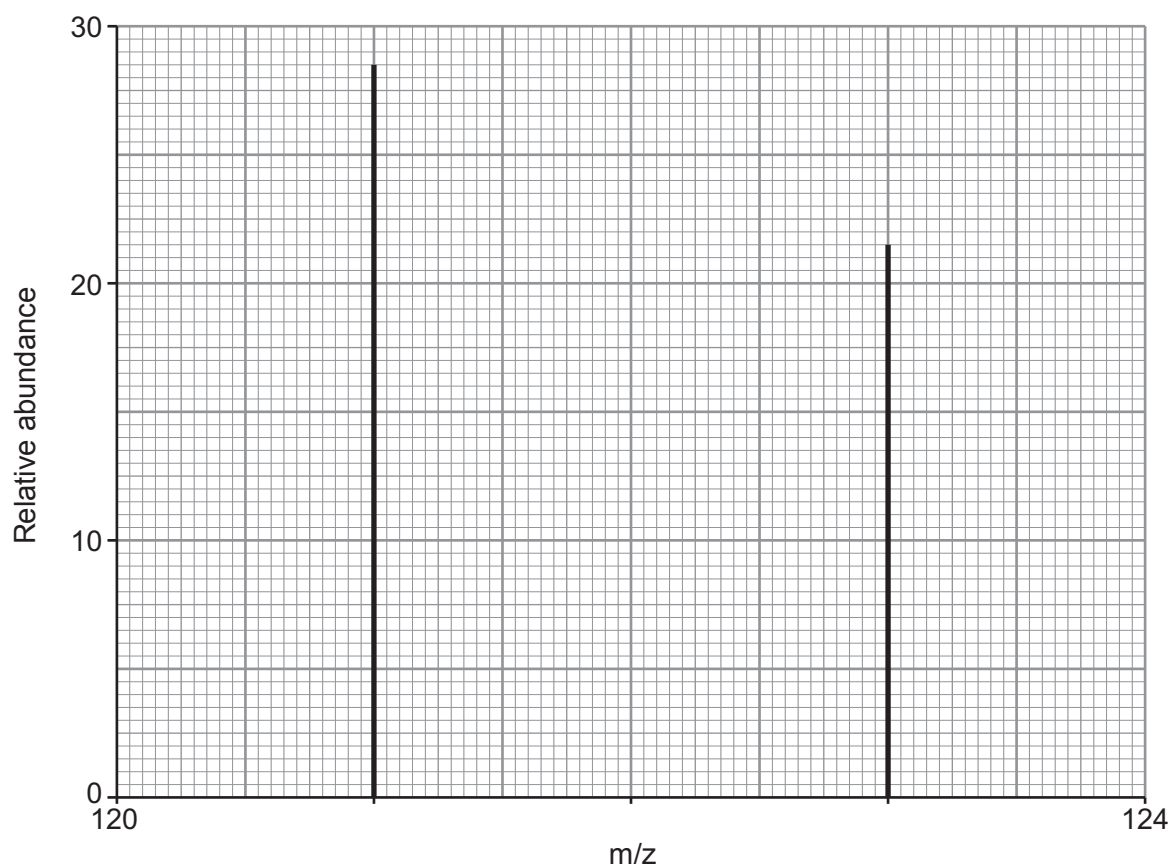


(c) Antimony is in Group 5. Its mass spectrum shows that it has two stable isotopes.



Calculate the relative atomic mass of antimony. You **must** show your working.

[2]

Relative atomic mass = .....



- (d) Although radiation from radioisotopes is harmful to health many beneficial uses of radioactivity have been found.

The table below gives some information about four radioactive isotopes.

| Isotope           | Radiation emitted | Half-life |
|-------------------|-------------------|-----------|
| $^{90}\text{Sr}$  | $\beta$           | 28 years  |
| $^{99}\text{Tc}$  | $\gamma$          | 6 hours   |
| $^{210}\text{At}$ | $\alpha$          | 8.1 hours |
| $^{228}\text{Th}$ | $\alpha$          | 1.9 years |

Use **all** the data given to choose which isotope is the most suitable to use as a tracer in medicine. Explain your answer. [3]

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